Space travel
A look at current and future plans to conquer the final frontier
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THE END OF THE WORLD AS WE KNOW IT?

Arguably one of the most exciting experiments in the history of science is about to resume, with the operation of the Large Hadron Collider in Switzerland. And yet some people are concerned that it could bring about the end of the world. We look at the main issues involved.

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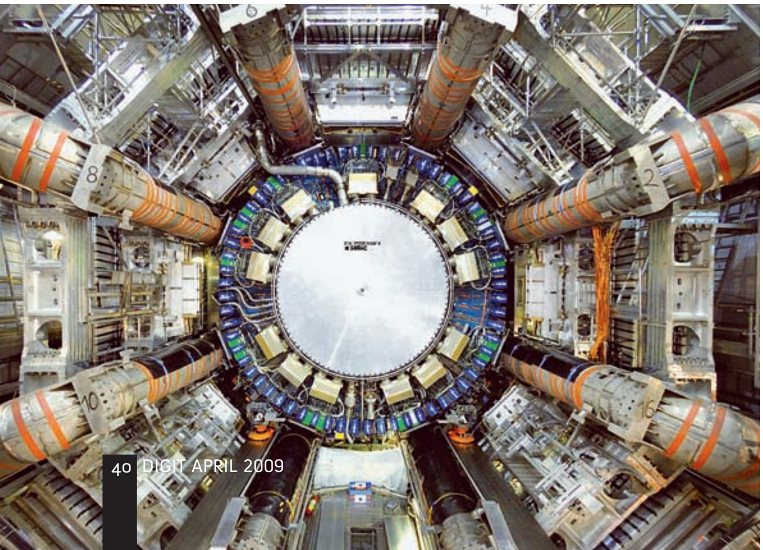
In May 1990, Tim Berners-Lee invented the system we now call the World Wide Web, in order to manage huge amounts of data across a large organisation. In his, and the world's, first web page (**www.w3.org/History/1989/proposal-msw.html**), he gave the motive for this project: "Many of the discussions of the future at CERN and the LHC era end with the question - 'Yes, but how will we ever keep track of such a large project?'" The LHC (Large Hadron Collider) era is now just about to start. After an unfortunate accidental damage last year, CERN is now expecting the LHC to produce its first beams at the end of September this year, and collisions late in the following month.

For any of you that have been on another planet for the last decade, the LHC is in Switzerland, and is the world's largest and most powerful particle accelerator. Of the particles concerned, hadrons, the best known are protons and neutrons, although the LHC will be able to accelerate much larger particles — lead nuclei are mentioned, for example. The purpose of the collider is to accelerate particles in two large near-circles, in opposite directions, very close to the speed of light, and collide them head on. These interactions produce other particles which are then detected and investigated.

The greater the energy of the collisions, the more varied and exotic are the types of particles that the accelerators produce, and the LHC is expected to reach the kind of energies that existed shortly after the Big Bang that is thought to have created the universe. You may have read that the LHC will create the conditions that existed shortly after the Big Bang. This is slightly misleading, because what is meant is that the energies and densities involved will be similar to those thought to have existed at that time. With the LHC, physicists are hoping to detect the existence of the illusive Higgs boson, thought to be responsible for the mass of all particles; magnetic monopoles are also a possibility; also, strangelets, and many others. This undeniably represents a very important set of experiments that could either support or refute our current theories of advanced physics. But among all this, the creation of one type of particle has particularly caught people's attention: a miniature black hole (BH).

The ATLAS detector at the LHC — a possible location for the creation of miniature black holes



There has been much discussion about both whether the LHC will produce miniature BHs at all, and the effects of these if it does. The worst scenario that has been described, goes like this. Because the particles that produce the BH are travelling in opposite directions, with very similar energies, when the BH is produced it has relatively low kinetic energy, and hence low velocity, and is caught by the Earth's gravitational field.

Very slowly at first — with this process perhaps taking years — the BH hits occasional particles and it absorbs them, each time gaining a little more mass and becoming a little larger. As this happens, it gradually circles around the centre of the Earth, getting nearer as it becomes more massive and slows down. Eventually the BH starts to absorb whole atoms, and its size increases more rapidly, until it is almost always in touch with molecules of the Earth's core, and absorbs them all.

A point is reached when the BH has absorbed so much matter, that is has created a small vacuum at the centre of the Earth, and matter continually falls into it. As this happens, the Earth's mantle contracts a little, causing the crust — the continents and ocean floor — to be compressed laterally, very slightly. This causes at first minor earthquakes; not only in places where earthquakes are common, but also in places where you would least expect them. This would almost certainly be the first sign that something had gone wrong.

After this point, the movement downwards of the core and mantle would produce enormous heat in the interior of the Earth, causing major volcanic eruptions, and ever more powerful earthquakes. This geologic activity would heat the oceans, eventually boiling them away, and the surface of the continents would be overrun with lava flows. The whole process would accelerate very quickly, with the whole Earth being sucked into the BH, perhaps within hours of the first major volcanic eruptions. All that would be left would the Moon and all our artificial satellites orbiting a point in space where the centre of the Earth used to be. So, just how likely is this?

The first point is clear — that nobody knows whether the LHC would be able to produce any BHs at all. If certain aspects of superstring theory are correct, then it might well do so. Much more important is the question of any dangers posed by such holes.

Naturally, the scientists at CERN have been keen to explain how safe is the LHC, but they have done a very bad job in PR terms. CERN reported that a risk assessment had been performed and concluded that the LHC was safe. But can we read the details of the report online? No. And the report was conducted by the safety group of the LHC — not a third-party independent group. It reminds one of the bankers who a couple of years ago were